

Bird dialect detector

Like humans birds too have dialects!



No



Yes

A slide playing two bird recordings

The hook

Goal : Determine where a bird is from, from its audio (song/calls)

- Pick one songbird species
- Take bird audio samples from different geographical regions of Europe
- Convert samples to Mel-spectrograms
- What is a Mel-spectrogram ?
 - A visual heat-map of sound frequencies over time, scaled to match human hearing.
- Why use them?
 - Models like CNNs excel at 2-D pattern recognition

Data processing

- Bird species : **Yellowhammer**
- Bird audio sample source :
Xeno-canto
- **Xeno-canto** : citizen science project and repository
- **Audio sample** : User submitted recordings with metadata like where it was recorded, quality , length,etc.



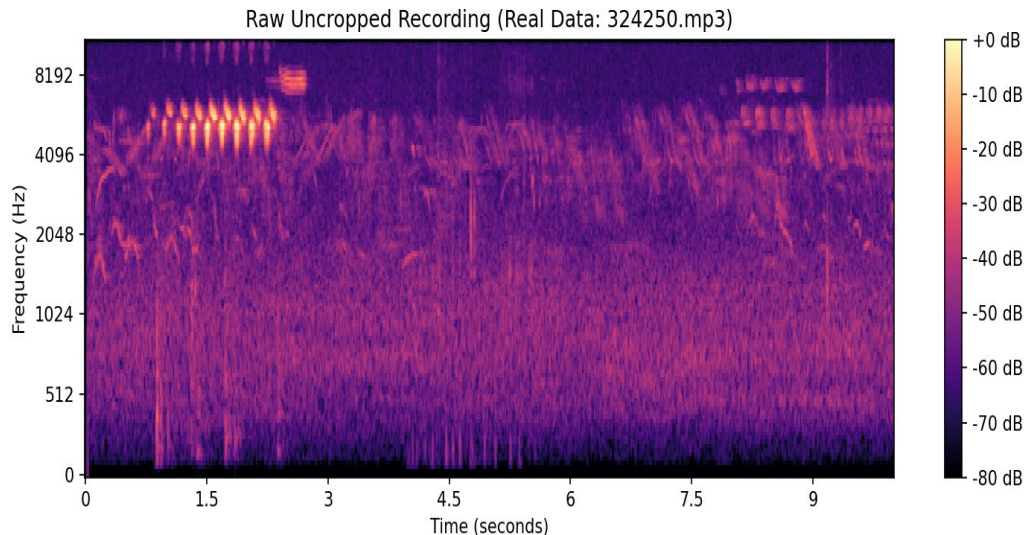
Data processing

Data cleanup

- Bird audio is present in only part of the sample
- Most of the bird song is the same. Need to extract and work on the part that is different.

How to get the different part ?

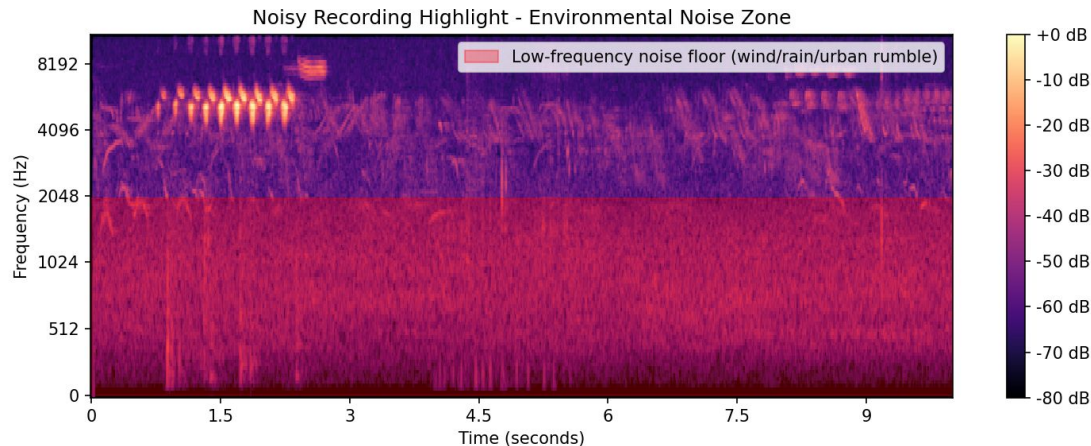
- A study highlighted that last 3 seconds of the song were distinct in yellowhammers



Data processing

Data cleanup

- Need to remove low frequencies to get water and ambient sound data out.



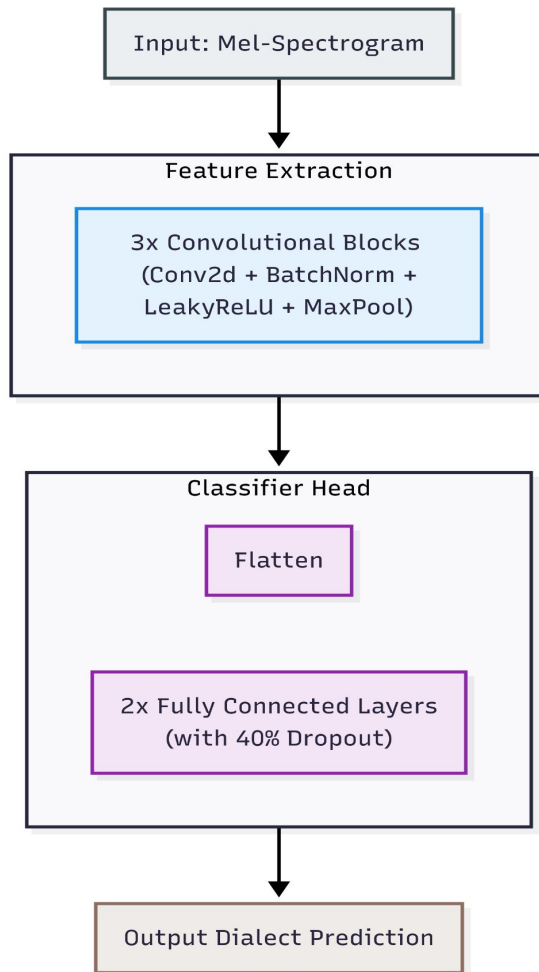
- Have to be careful about not potentially cropping out the parts that might be unique to the bird.
- With low domain knowledge , hard to determine.

Model

INPUT

: 40 samples per region/class

- 32 to train
- 8 to evaluate



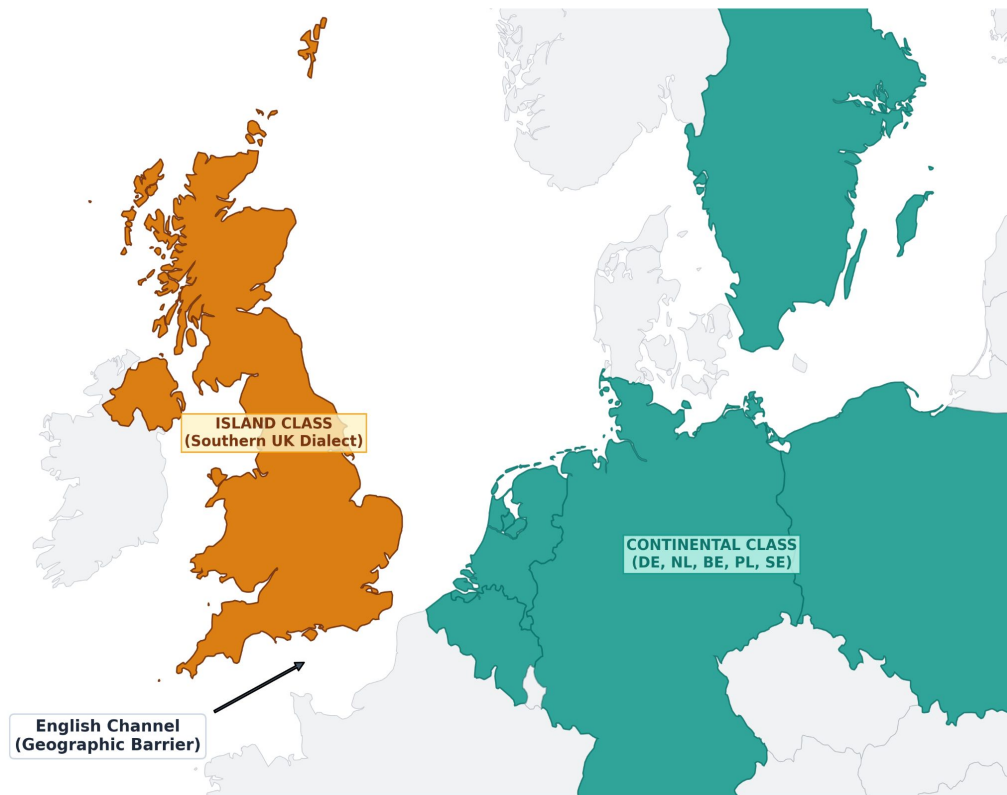
Results

- Literature says bird dialects are hard to classify as its ~ 95% the same.
- 5 regions : southern uk, east germany , netherlands + belgium , south poland , southern sweden
- The literature was right.
- In the 5 regions, **~40%** accuracy — barely above the 20% random baseline

Results

- Retrained as a Binary Classifier:
Island (UK) vs Continent
- **87.5 %** accuracy
- The English Channel is the
real dialect boundary!

Yellowhammer Song Dialects: Geographic Experiment Layout



Island biogeography

- Geographic isolation accelerates divergence of traits (like bird song dialect)
- Yellowhammers are sedentary birds. Don't move around much.
- The English Channel acts as a physical barrier and distinct patterns are developed.



Further work

- Testing this pattern with one other sedentary bird :
Chaffinch
- Trying to build a reusable tool
- Write a blog post , open source and try to reach out to people in this space who might find this interesting.

THE END

